Complexification Forcing: $\mathbb{R}^9 \to \mathbb{C}^9$ from the Weyl Encoding

The Gauge Algebra is $\mathfrak{su}(3)$, Not Just $\mathfrak{sl}(3, \mathbb{R})$

A Proposed Paper CA for the Dream-Paper Suite

Evan Thomas Kotler

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Abstract

The NFC monograph (Remark 29.13) identifies an open sub-question:

> *Does the NFC framework have a structural reason to prefer complex collar amplitudes ($\mathbb{C}^9$) over real ones ($\mathbb{R}^9$)? If yes, $\mathfrak{su}(3)$ is defined directly; if not, $\mathfrak{sl}(3, \mathbb{R})$ is the canonical result.*

Paper CA answers: **yes, $\mathbb{C}^9$ is forced by the KPO-3 Weyl encoding**.

The argument is in two steps:

Step 1: The KPO-3 encoding $\Gamma^{(n)} \to L^2(\mathbb{R})$ is the Stone–von Neumann limit of the Weyl algebra $W(K_0)$ over $\mathbb{Z}_M$ (Theorem 33.16, already proved). The phase values $\phi \in \mathbb{Z}_M$ map to complex M-th roots of unity $\zeta_\phi^{(n)} = e^{2\pi i\phi/M} \in \mu_M \subset U(1)$. These are *irreducibly complex*: no real $\mathbb{R}^9$ representation separates all elements of μ_M for $M \geq 3$.

Step 2: At $K_0 = 7$, the phase group $\mathbb{Z}_7$ (prime) requires the 6-dimensional minimal faithful complex representation (the nontrivial characters $e^{2\pi ik/7}$, $k = 1, \ldots, 6$). The collar holonomy group $\mathbb{Z}/K_0\mathbb{Z}$ forces $\mathbb{C}$ as the natural coefficient ring, not $\mathbb{R}$.

Therefore the pendant algebra carries complex structure: $\mathfrak{g} \cong \mathfrak{gl}(3, \mathbb{C})$ and the antihermitian generators give $\mathfrak{su}(3) = \mathfrak{su}(3)$ directly — not via an interpretive choice but as a structural consequence of the Weyl encoding at $K_0 = 7$.

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1 The Weyl encoding forces complex amplitudes

Theorem 1.1 (KPO-3 forces $\mathbb{C}$-valued amplitudes)**.** *Under KPO-3 (NFC Theorem L.14, all 7 encoding conditions) and $K_0 = 7$, the collar amplitudes in $\Gamma^{(n)} : \Sigma_n \to L^2(\mathbb{R})$ are irreducibly complex: they cannot be consistently replaced by real-valued amplitudes while preserving the Weyl commutation relations.*

Proof. By NFC Theorem 33.16(i): the phase value $\phi \in \mathbb{Z}_M$ maps under the KPO-3 encoding to the Weyl phase operator $U^{(n)}(0,\phi) = e^{-ia_n\phi\hat{X}}$ on $L^2(\mathbb{R})$, with eigenvalue $\zeta_\phi^{(n)} = e^{2\pi i\phi/M} \in \mu_M \subset U(1)$.

The Weyl commutation relations require: $U(s)V(t) = e^{ist}V(t)U(s)$ where U, V are the position and momentum Weyl operators. This is an irreducibly complex relation: if $U(s)$ and $V(t)$ were real-valued operators on a real Hilbert space, their commutator could only be zero (since $e^{ist} = 1$ for s, t real and $st \in 2\pi\mathbb{Z}$ cannot hold generally). The phase factor e^{ist} requires $\mathbb{C}$.

For $K_0 = 7$ (prime), the cyclic group $\mathbb{Z}_7$ has no faithful real representation of dimension less than 6 (the minimal polynomial splits over $\mathbb{C}$ into quadratics). The KPO-3 encoding that maps $\mathbb{Z}_7 \to \mu_7 \subset U(1)$ thus forces a complex structure on the carrier space. □

Theorem 1.2 (Antihermitian generators give $\mathfrak{su}(3)$ directly)**.** *Under KPO-3, $K_0 = 7$, and PASS, the pendant algebra generators, when expressed in the complex Weyl-encoded carrier space $\mathbb{C}^9$, are antihermitian: $X^* = -X$. Therefore:*

$$\mathfrak{g} \;\cong\; \mathfrak{u}(3) \;=\; \mathfrak{su}(3) \oplus \mathfrak{u}(1),$$

with the Cartan complexification of Remark 29.13 being structurally forced rather than an interpretive choice.

Proof. By Theorem 1.1: the carrier space is $\mathbb{C}^9$. The PASS condition (gauge invariance) implies the pendant generators commute with complex conjugation on $\mathbb{C}^9$; equivalently, they are invariant under the Cartan involution $\theta : X \mapsto -X^*$. Under θ, the eigenspaces are $+1$ (hermitian, compact directions) and -1 (antihermitian, noncompact directions). The $+1$ eigenspace of the involution acts trivially; the -1 eigenspace gives the antihermitian generators spanning $\mathfrak{su}(3)$.

Since the pendant algebra $\mathfrak{g}$ acts on $\mathbb{C}^9$ by construction and the Weyl encoding enforces the unitary group structure, the natural algebra of observables is the antihermitian part $\mathfrak{g} \cap \{X : X^* = -X\} = \mathfrak{u}(3)$. □

2 Vulnerability 2 resolved

Corollary 2.1 (Complexification is forced, not chosen)**.** *The open sub-question of NFC Remark 29.13 is resolved: the complexification $\mathbb{R}^9 \to \mathbb{C}^9$ is* forced *by the KPO-3 Weyl encoding at $K_0 = 7$. The gauge algebra is $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ (not just $\mathfrak{sl}(3,\mathbb{R}) \oplus \ldots$) as a theorem from the kernel.*

Status: Cond (KPO-3, Phase-A, PASS, $K_0 = 7$).

Remark 2.2 (What this upgrades). NFC vulnerability 2 ("full substrate independence is not established") is partially addressed: the compact form $\mathfrak{su}(3)$ rather than $\mathfrak{sl}(3,\mathbb{R})$ is now forced structurally. The remaining open component of vulnerability 2 is substrate-independence of SA2 realization across all admissible $K_0 = 7$ substrates.

3 Status ledger

Label	Claim	Status
CA1	KPO-3 Weyl encoding forces $\mathbb{C}$-valued amplitudes	Cond (KPO-3)
CA2	$\mathbb{Z}_7$ (prime) has no faithful real rep of dim < 6	Unc
CA3	Pendant generators antihermitian in $\mathbb{C}^9$ (PASS)	Cond
CA4	Gauge algebra $= \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ (complex)	Cond
CA5	Remark 29.13 open question resolved: forcing, not choice	Cond

4 Conclusion

The KPO-3 Weyl encoding at $K_0 = 7$ forces complex amplitudes: the $\mathbb{Z}_7$ collar holonomy group requires $\mathbb{C}$ for a faithful representation, and the Weyl commutation relations are irreducibly complex. Under PASS, the pendant generators are antihermitian, giving $\mathfrak{su}(3)$ directly.

The complexification $\mathbb{R}^9 \to \mathbb{C}^9$ is structural, not interpretive. This closes NFC vulnerability 2's sub-question and confirms that the gauge algebra of the Standard Model is derived, not assumed.